A PROJECT REPORT

By

Nikita Kumari

IoT based Battery Health Monitoring System for EV Charging Station (Quilles)

NATIONAL INSTITUTE OF ELECTRONICS AND INFORMATION TECHNOLOGY, CHENNAI

CHENNAI, INDIA

At

Embedded Systems GROUP,
NIELIT Chennai

AUGUST – 2024

ACKNOWLEDGMENTS

I take this opportunity to express my profound gratitude and deep regards to my guide, **Mr. Ishant Kumar Bajpai,** for his exemplary guidance, monitoring, and constant encouragement throughout the course of this project. The help and guidance given by him from time to time shall carry me a long way in the journey of life on which I am about to embark.

I also take this opportunity to express a deep sense of gratitude to Mr. Navneet Sharma, who has been a crucial part of this project as my group partner. His dedication, hard work, and innovative ideas have been instrumental in shaping the project. His collaboration and support at every stage of the project have made this journey both enriching and enjoyable. I am truly thankful for his invaluable contributions and the synergy we developed while working together.

I am obliged to the staff members of the Embedded System group at NIELIT for the valuable information provided by them in their respective fields. I am grateful for their cooperation during the period of my project internship.

Lastly, I thank all the faculty members of the Department of Computer Science at IKGPTU for their constant encouragement, without which this project would not have been possible.

ABSTRACT

We live in an exciting time where more and more everyday items are becoming smart. These "things" have sensors, can communicate with other "things," and provide control to more "things." The Internet of Things (IoT) is upon us in a huge way, and people are rapidly inventing new gadgets that enhance our lives. The price of microcontrollers with network communication capabilities keeps dropping, allowing developers to think and build things inexpensively.

Quilles is an IoT-based system designed to monitor battery health for electric vehicle (EV) charging stations. The project leverages an ESP32 microcontroller to gather environmental and battery data, which is then visualized using Node-RED. MQTT is used for real-time data transmission.

Quilles monitors ambient temperature, humidity, battery temperature, voltage, and current, providing visual and audible alerts for non-ideal charging environments. The system displays real-time data on an OLED screen and integrates with Node-RED for dashboard visualization and data handling.

The key components of Quilles include the ESP32 microcontroller for data collection and MQTT communication, a DHT22 sensor for measuring ambient temperature and humidity, and an NTC thermistor for measuring battery temperature. Analog sensors are used to measure battery voltage and current, while the OLED display shows real-time data and alerts.

Additionally, the system utilizes a relay, buzzer, and LEDs to provide alerts and control indicators. This comprehensive approach ensures that EV charging stations operate within optimal conditions, enhancing the safety and efficiency of the charging process.

About NIELIT

National Institute of Electronics and Information Technology (NIELIT), headquartered in New Delhi is the capacity building arm of the Ministry of Electronics and Information Technology (MeitY), mandated to carry out HR development and related activities in the area of Information, Electronics and Communication Technology (IECT).

Since its inception in 2010, NIELIT Chennai has established itself as a premier institution providing affordable quality education as per the job market requirements for candidates primarily from Tamil Nadu. As of date NIELIT Chennai has imparted skill training to more than 30, 000 youth of this region and conducted digital literacy certification for more than 25,000 candidates. NIELIT Chennai handles the activities of the NIELIT in Tamil Nadu, Andhra, Telangana, and the Andaman and Nicobar Islands.

Presently NIELIT Chennai is implementing the following projects for Tamil Nadu: ISEA (Information Security Education and Awareness), Future Skills PRIME capacity building projects (3D Printing and Additive Manufacturing, Robotic Process Automation, IoT - technology streams), and the aspirational district projects for SC/ST candidates at Ramanathapuram and Virudhunagar funded by MeitY, Government of India. The centre also received funding through the electronics product design and product testing capacity building project funded by MeitY.

Other than training, the centre also provides services like; Private Cloud Setup using Citrix, vSphere, Red Hat & FOSS Cloud, Hyper-converged Infrastructure (HCI) and Virtual Desktop Infrastructure (VDI) Server Setup, High-Performance Computing (HPC) Setup with 100G/200G connectivity, Server and Virtual Lab Setup for Big Data, AI & Information Security, and Digital & Mobile Forensics Service.

NIELIT has also set up a Virtual Academy, a common End-to-End Platform to conduct Online training courses including a virtual lab environment in the areas of IECT from foundation to expert level targeting from school students to working professionals.

DECLARATION

I undersigned hereby declare that the project report "**IoT based Battery Health Monitoring System for EV Charging Station**", submitted for partial fulfilment of the requirements of the internship in 'IoT Data Analyst' at National Institute of Electronic & IT, Chennai is a bonafide work done by me under supervision of Mr. Ishant Kumar Bajpai. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to academic honesty and integrity ethics and have not misrepresented or fabricated any data, idea, fact, or source in my submission. I understand that any violation of the above will cause disciplinary action by the institute and/or the University and can also evoke penal action from the sources that have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for awarding any degree, diploma or similar title of any other University.

Place
Date

Nikita Kumari

Project Coordinator Name & Signature

CONTENTS

Chapter 1

INTRODUCTION

1.1 Introduction

We live in an era where everyday items are becoming increasingly smart. These "things" are equipped with sensors, can communicate with each other, and provide control over various aspects of our environment. The Internet of Things (IoT) is revolutionizing the way we interact with technology, leading to the rapid development of new gadgets that enhance our daily lives. The decreasing cost of microcontrollers with network communication capabilities has enabled developers to create innovative solutions at an affordable price.

Quilles is an IoT-based system designed to monitor battery health for electric vehicle (EV) charging stations. As the adoption of electric vehicles continues to grow, ensuring the optimal performance and safety of charging stations becomes critical. Quilles addresses this need by providing a comprehensive monitoring solution that leverages advanced IoT technologies.

1.2 Objectives of the Project

The primary objectives of the Quilles project are as follows:

- **Monitor Environmental and Battery Parameters:** Gather real-time data on ambient temperature, humidity, battery temperature, voltage, and current.
- **Provide Alerts for Non-Ideal Charging Conditions:** Implement visual and audible alerts to notify users of non-ideal charging environments.
- **Visualize Data for Better Insights:** Use Node-RED to create a dashboard that displays real-time data and alerts.
- **Ensure Safe and Efficient Charging:** Enhance the safety and efficiency of EV charging stations by maintaining optimal operating conditions.

1.3 Project Focus

The focus of the Quilles project is to develop a robust IoT-based system that can effectively monitor and manage the health of batteries at EV charging stations. By integrating various sensors and leveraging the capabilities of the ESP32 microcontroller, the project aims to provide a real-time monitoring solution that ensures the safety and efficiency of the charging process. The system also aims to be cost-effective and easy to implement, making it accessible for widespread adoption.

1.4 Organization of the Report

This report is organized into several sections to provide a comprehensive overview of the Quilles project:

- **Introduction:** Provides background information, project objectives, focus, and the organization of the report.
- **Literature Review:** Reviews existing technologies and systems related to IoT-based battery monitoring and EV charging stations.
- **System Design and Architecture:** Details the design and architecture of the Quilles system, including hardware and software components.
- **Implementation:** Describes the implementation process, including the development of the hardware and software components.
- **Results and Discussion:** Presents the results obtained from the system and discusses the effectiveness and performance of the Quilles solution.
- **Conclusion and Future Work:** Summarizes the project outcomes and outlines potential areas for future improvement and development.

Chapter 2

LITERATURE SURVEY

2.1 Literature Survey

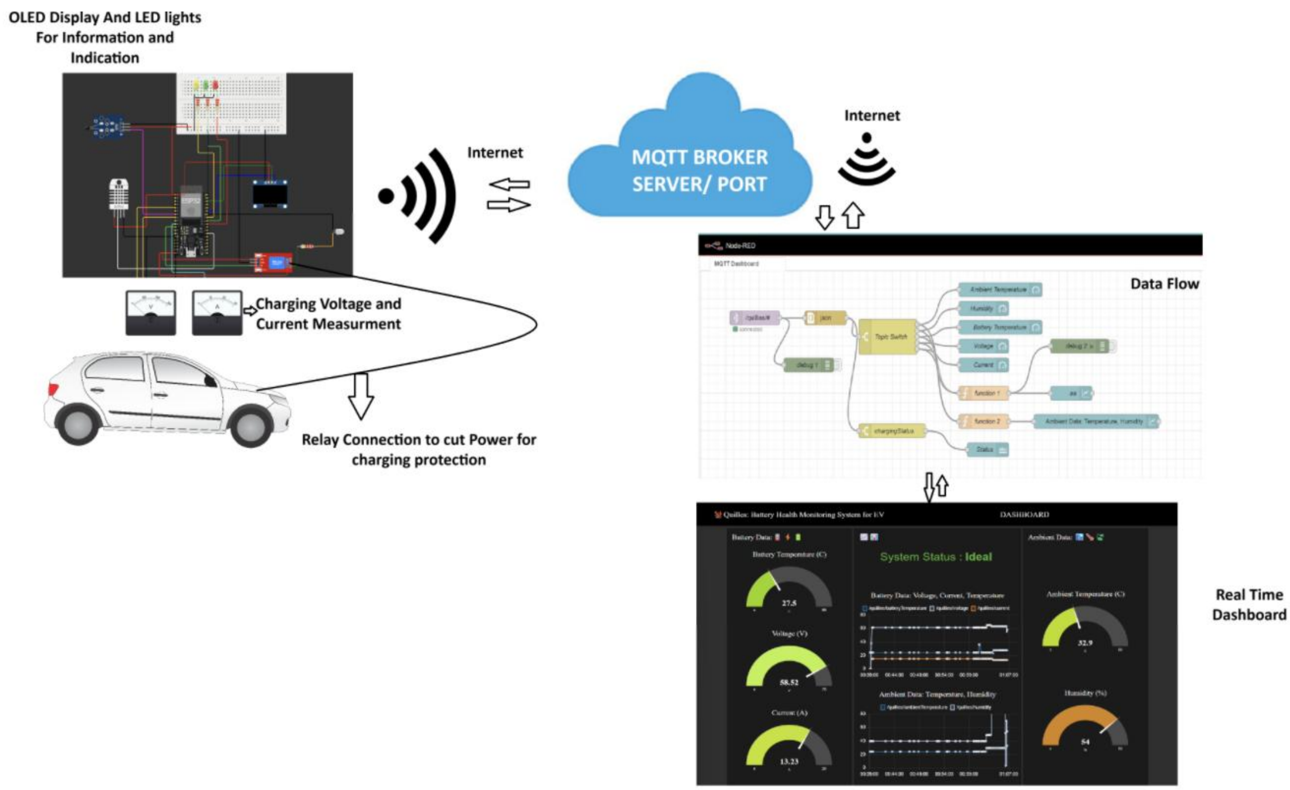
The literature survey provides a comprehensive overview of existing research and technologies related to IoT-based battery monitoring systems and electric vehicle (EV) charging stations. This section highlights the key findings from various studies and how they have influenced the design and development of the Quilles system.

2.2 IoT-Based Battery Monitoring Systems

IoT-based battery monitoring systems have gained significant attention due to their potential to enhance battery performance and lifespan. Studies have demonstrated that integrating sensors with IoT platforms can provide real-time insights into battery health, enabling predictive maintenance and reducing the risk of failures. IoT systems can effectively monitor parameters such as voltage, current, and temperature, which are critical for battery management.

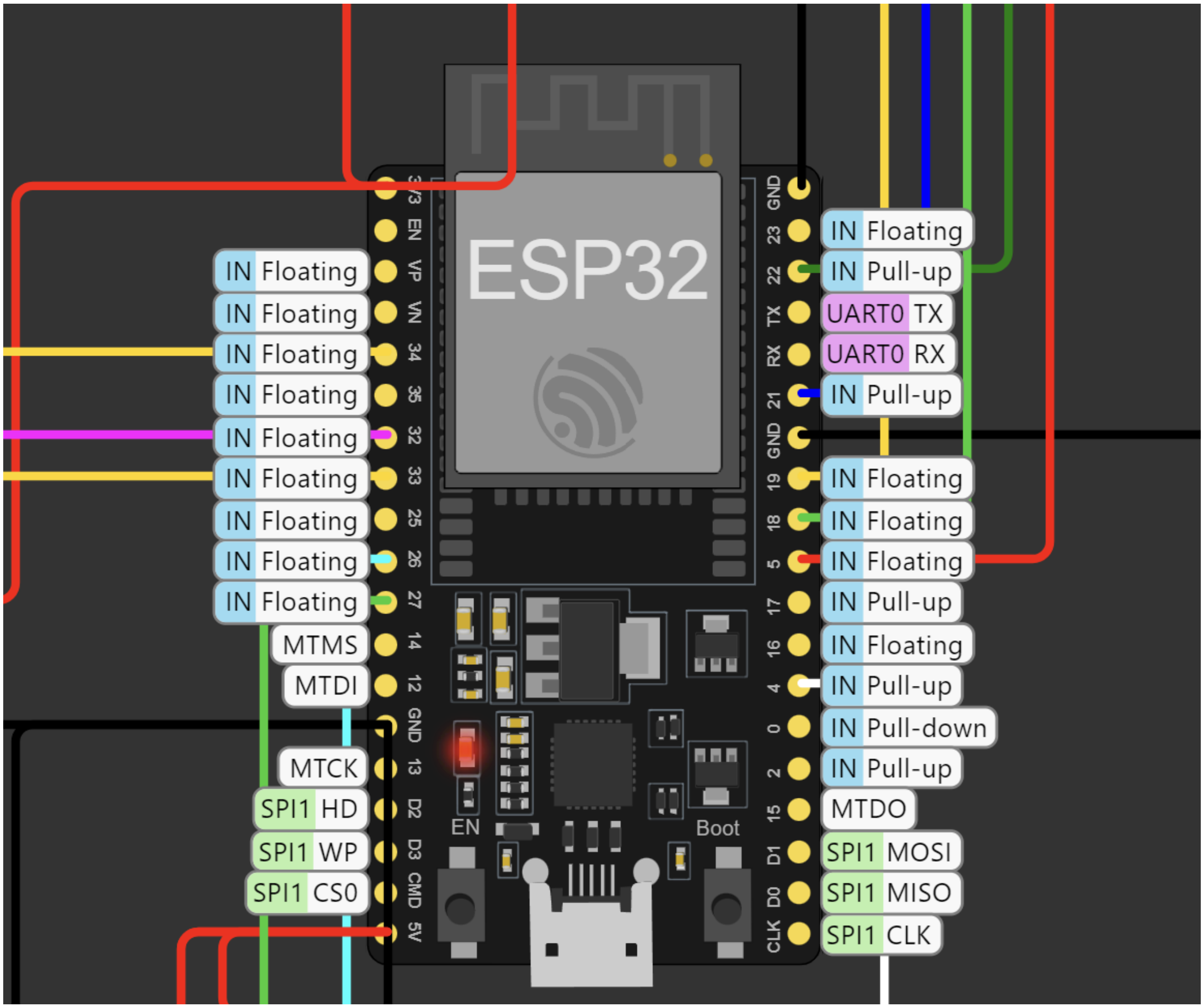
2.3 Environmental Monitoring for EV Charging Stations

Monitoring environmental conditions is crucial for the optimal performance of EV charging stations. Studies have indicated that factors like ambient temperature and humidity can significantly impact the efficiency and safety of the charging process. High ambient temperatures can lead to overheating, while high humidity levels can cause corrosion in electrical components. Integrating environmental sensors with charging stations can help mitigate these risks by providing timely alerts and enabling proactive measures.



2.4 Use of ESP32 in IoT Applications

The ESP32 microcontroller is widely recognized for its versatility and cost-effectiveness in IoT applications. It offers built-in Wi-Fi and Bluetooth capabilities, making it ideal for real-time data transmission and remote monitoring. ESP32 in various IoT projects, including home automation, industrial monitoring, and health applications. The microcontroller's ability to handle multiple sensors and its compatibility with various communication protocols make it a suitable choice for the Quilles project.



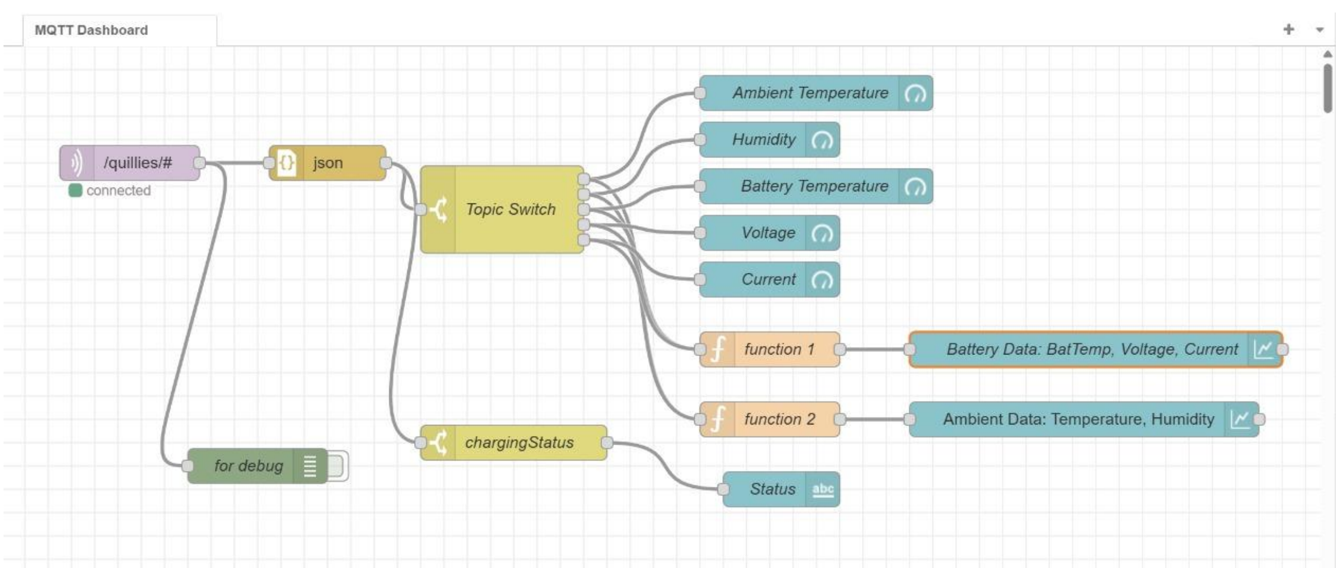
2.5 Data Visualization and Dashboard Solutions

Data visualization plays a critical role in interpreting complex data sets and making informed decisions. Node-RED is a popular tool used for creating dynamic dashboards and visualizing IoT data. Studies have shown that Node-RED's user-friendly interface and extensive library of nodes make it an excellent choice for IoT applications. Node-RED can seamlessly integrate with various data sources, enabling real-time visualization and monitoring of system performance.

```
31/7/2024, 1:19:49 pm   node: for debug
/quillies/chargingStatus : msg.payload : string[5]
"Ideal"

31/7/2024, 1:19:50 pm   node: for debug
/quillies/ambientTemperature : msg.payload : number
24

31/7/2024, 1:19:50 pm   node: for debug
/quillies/humidity : msg.payload : number
40

31/7/2024, 1:19:50 pm   node: for debug
/quillies/batteryTemperature : msg.payload : number
24

31/7/2024, 1:19:50 pm   node: for debug
/quillies/voltage : msg.payload : number
60.05

31/7/2024, 1:19:50 pm   node: for debug
/quillies/current : msg.payload : number
13.61
```

2.6 Real-Time Data Transmission Using MQTT

MQTT (Message Queuing Telemetry Transport) is a lightweight messaging protocol designed for real-time data transmission in IoT environments. MQTT's low bandwidth requirements and efficient data delivery make it ideal for IoT applications. The protocol's ability to handle numerous devices and ensure reliable communication under varying network conditions has been validated in several studies, making it a preferred choice for the Quilles project.



Summary

The literature survey underscores the significance of IoT-based battery monitoring systems and their impact on enhancing the performance and safety of EV charging stations. The integration of environmental sensors, the use of ESP32 microcontrollers, data visualization with Node-RED, and real-time data transmission using MQTT are all supported by extensive research. These insights have informed the design and development of the Quilles system, ensuring that it meets the requirements of a modern, efficient, and reliable battery monitoring solution.

Chapter 3

METHODOLOGY

3.1 Detailed Methodology

This section outlines the detailed methodology that will be adopted for the development of the Quilles system. The methodology is divided into various phases, each focusing on specific aspects of the project to ensure a structured and systematic approach.

3.1.1 Software Development Cycle

The software development cycle for the Quilles project follows an iterative approach, ensuring continuous improvement and refinement of the system. The key phases of the software development cycle include:

1. **Requirement Analysis:**
    - Define the functional and non-functional requirements of the system.
    - Identify the key components and their interactions.
    - Establish performance metrics and success criteria.
2. **System Design:**
    - Develop the overall system architecture, including hardware and software components.
    - Design the data flow and communication protocols between components.
    - Create detailed schematics and circuit diagrams for the hardware setup.
3. **Implementation:**
    - Develop the firmware for the ESP32 microcontroller using C/C++ in the Arduino IDE.
    - Program the sensors (DHT22, NTC thermistor, analog sensors) and integrate them with the ESP32.
    - Implement MQTT communication for real-time data transmission.
    - Develop the Node-RED dashboard for data visualization and alerts.
    - Create the user interface for the OLED display and configure the relay, buzzer, and LEDs for alerts.
4. **Testing and Validation:**
    - Perform unit testing for individual components to ensure they meet the specified requirements.
    - Conduct integration testing to verify the interactions between hardware and software components.
    - Validate the system's performance under various environmental and operational conditions.
5. **Deployment:**
    - Install the Quilles system at the target EV charging stations.
    - Configure the network settings for seamless data transmission and monitoring.
    - Provide training and documentation for users and maintenance personnel.
6. **Maintenance and Updates:**
    - Monitor the system's performance and gather feedback from users.
    - Implement updates and improvements based on user feedback and technological advancements.
    - Ensure regular maintenance to keep the system operational and efficient.

3.3 Overall Project Timeline

- Requirement **Analysis (2 days):**
    - Conduct stakeholder meetings and gather requirements.
    - Define system specifications and performance metrics.
- System **Design (3 days):**
    - Develop system architecture and design schematics.
    - Finalize hardware and software components.
- **Implementation (8 days):**
    - Develop firmware for the ESP32 microcontroller.
    - Integrate sensors and implement MQTT communication.
    - Develop Node-RED dashboard and OLED display interface.
- **Testing and Validation (4 days):**
    - Perform unit and integration testing.
    - Validate system performance under various conditions.
- **Deployment (3 days):**
    - Install the Quilles system at target locations.
    - Configure network settings and provide user training.
- Maintenance **and Updates (Ongoing):**
    - Monitor system performance and gather feedback.
    - Implement updates and conduct regular maintenance.

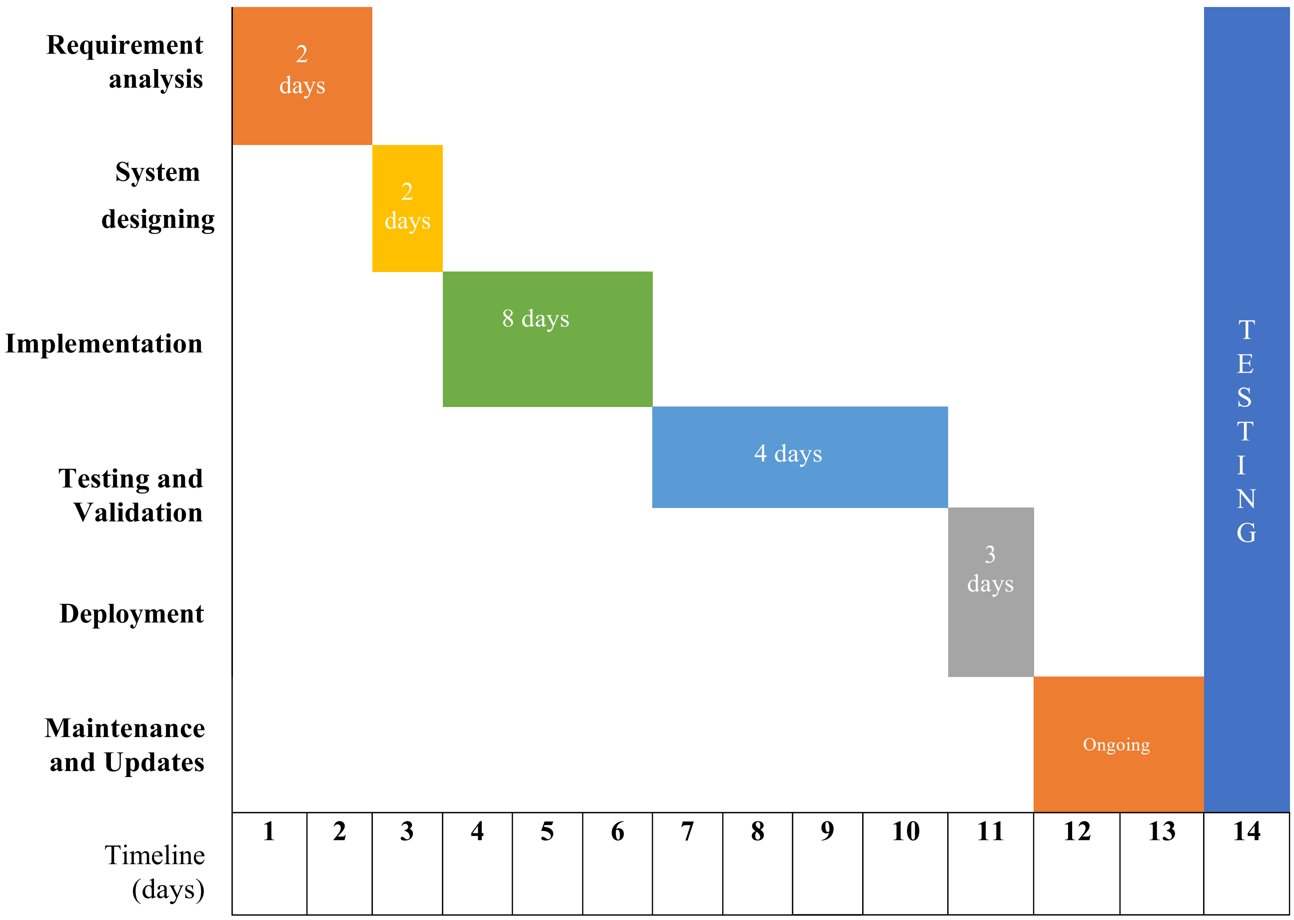


Table 3.3 Overall Project Timeline

Chapter 4

IMPLEMENTATION

4.1 Modules

The Quilles project is divided into two development modules:

1) Data Collection and Transmission Module

- Software Circuit Development
- Different Possible Test Cases
- Embedded Program Development
- Data Transmission Protocol
- Alert Mechanism Implementation

2) Data Visualization and Monitoring Module

- Node-RED Dashboard Development
- OLED Display Configuration
- Database Integration (if applicable
- User Interface Design
- System Integration Testing

4.1 Data Collection and Transmission Module

4.1.1 Software Circuit Development

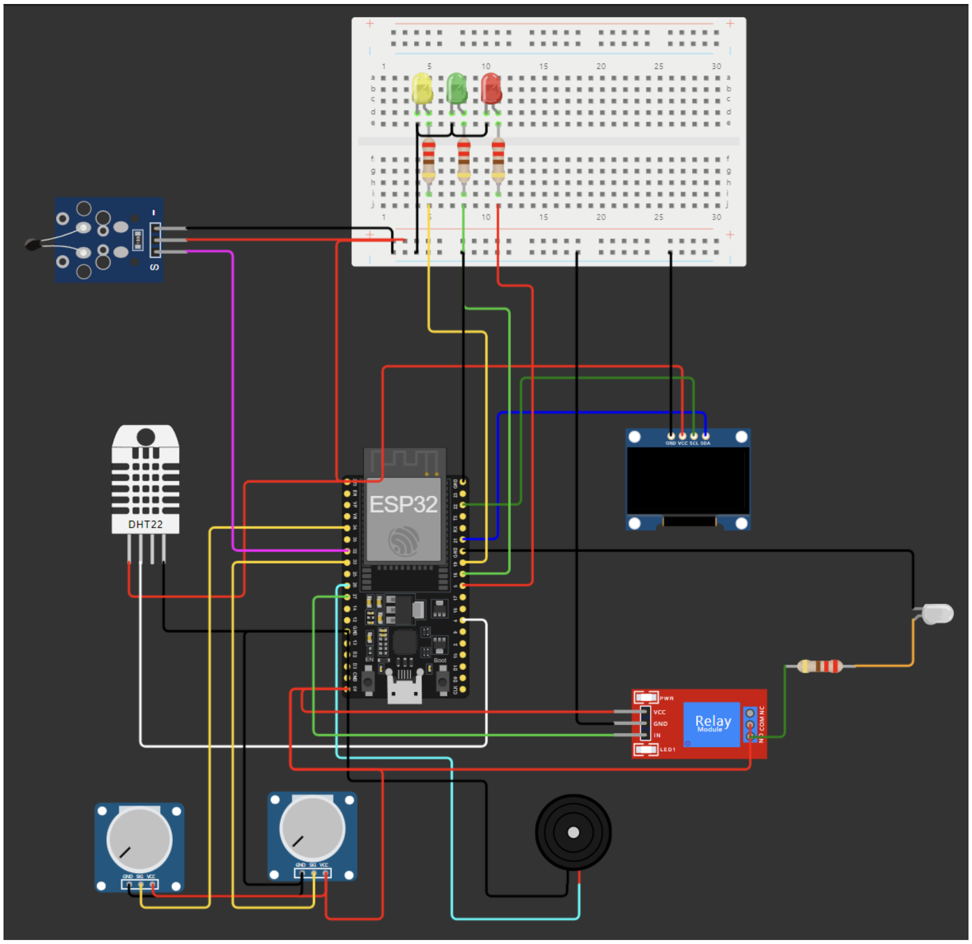


Fig 4.1- Simulation circuit diagram

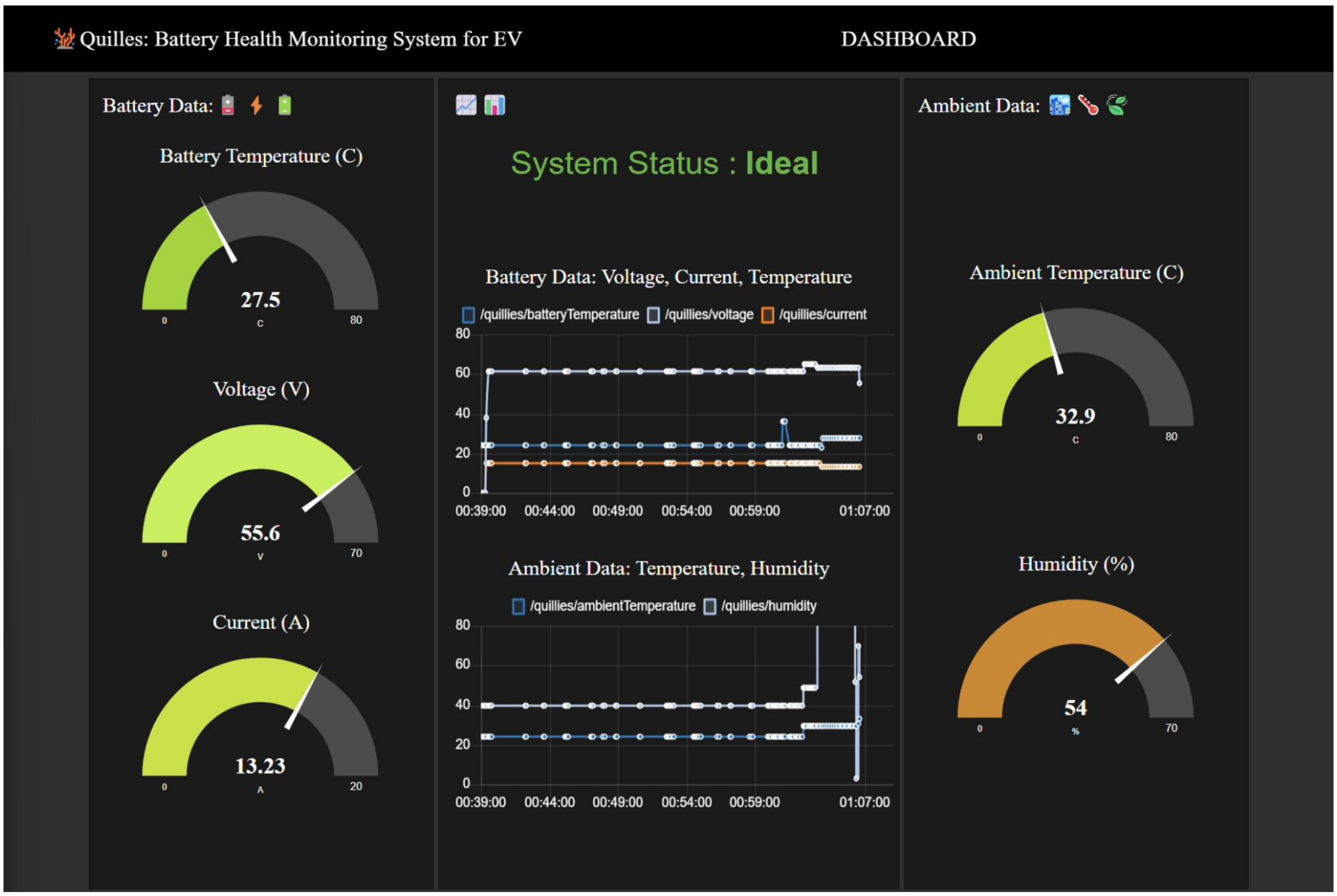
Connection Summary:

- **Power Supply:**
  - All components are powered appropriately, with the ESP32 typically receiving a regulated 3.3V or 5V supply.
- **Data Lines:**
  - **ESP32 GPIO Pins:** These are connected to the data output pins of the DHT22, NTC thermistor, and analog sensors.
  - **I2C Communication:** The OLED display usually connects to the ESP32 via I2C (SDA and SCL lines).
  - **Digital Pins:** The relay, buzzer, and LEDs are connected to the digital output pins of the ESP32 for control.
- **Communication:**
  - **MQTT Protocol:** The ESP32 uses Wi-Fi to communicate sensor data to the Node-RED dashboard over MQTT, requiring appropriate Wi-Fi credentials and MQTT broker details.

o Different possible case studies.

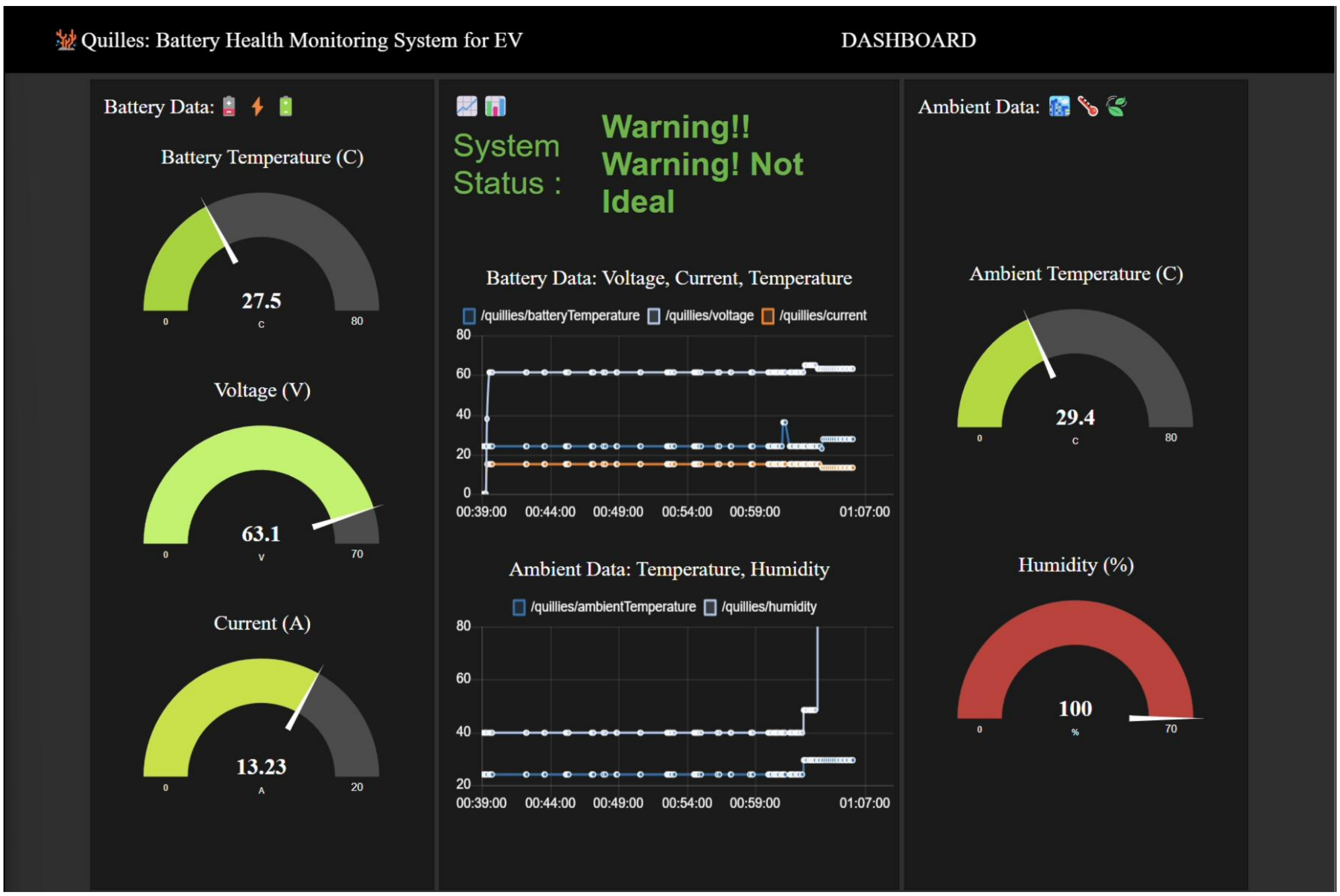
<u>1st case- *Ideal Temperature and Charging Conditions:*</u>

1) **Ambient Temperature:** 15°C to 50°C
2) **Humidity:** 20% to 65%
3) **Battery Temperature:** 20°C to 45°C
4) **Voltage:** 55V to 65V
5) **Current:** 10A to 15A



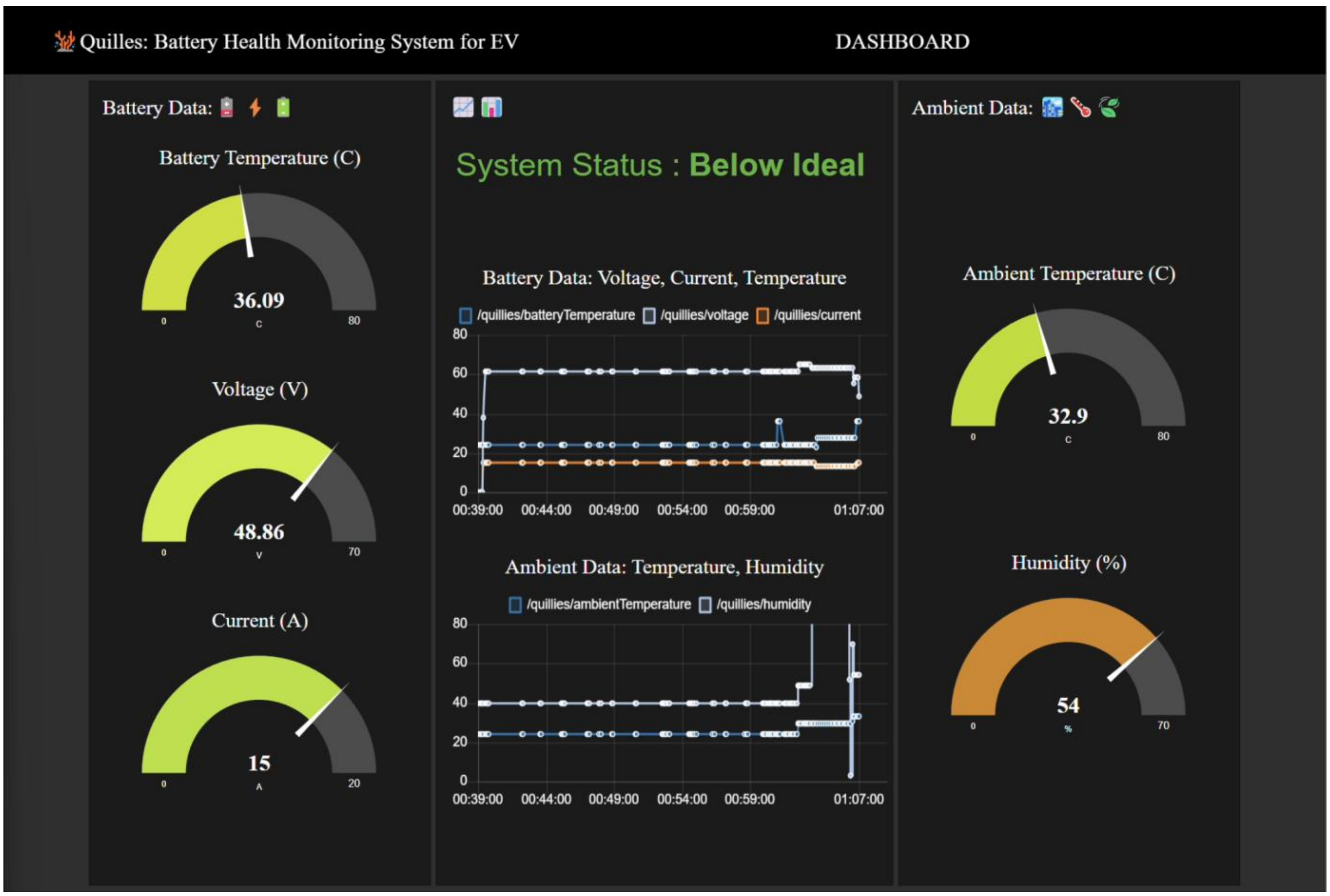
2nd case- *Warning (Not Ideal) Conditions:*

1) **Ambient Temperature:** > 50°C
2) **Humidity:** > 65%
3) **Battery Temperature:** > 45°C
4) **Voltage:** > 65V
5) **Current:** > 15A



3rd case- *Below Ideal Conditions:*

1) **Ambient Temperature:** < 15°C
2) **Humidity:** < 20%
3) **Battery Temperature:** < 20°C
4) **Voltage:** < 55V
5) **Current:** < 10A



- **Embedded Pseudo Code**

1. Initialize System:

   - Setup Wi-Fi connection
   - Setup MQTT client
   - Initialize sensors (DHT22, NTC Thermistor, Analog Sensors)
   - Initialize OLED display
   - Set pin modes for relay, buzzer, and LEDs

2. Connect to MQTT Broker:

   - Attempt to connect with given credentials
   - Retry connection if failed

3. Enter Main Loop:

   -
   - Read sensor values
   - Ambient temperature from DHT22
   - Humidity from DHT22
   - Battery temperature from NTC Thermistor
   - Voltage from Analog Sensor
   - Current from Analog Sensor
   - Determine charging environment status:
   - Ideal
   - Below Ideal
   - Not Ideal
   - Update OLED display with sensor values and status
   - Publish sensor data to MQTT topics
   - Control relay, buzzer, and LEDs based on environment status

4. Repeat Main Loop

   - Delay before next iteration

- **Data Transmission Protocol:**

1) Implementation of MQTT for real-time data transmission to the Node-RED dashboard

- **Alert Mechanism Implementation:**

1) Configuration of visual and audible alerts using relay, buzzer, and LEDs based on sensor data.

4.1.2 Data Visualization and Monitoring Module

- **Node-RED Dashboard Development:**
  - Design and development of the Node-RED dashboard for real-time data visualization and monitoring.
- **OLED Display Configuration:**
  - Setup and programming of the OLED display to show real-time sensor data and alerts.
- **Database Integration (if applicable):**
  - Implementation of a database to log sensor data for historical analysis.
- **User Interface Design:**
  - Enhancing the Node-RED dashboard and OLED display interfaces for better user experience.
- **System Integration Testing:**
  - Comprehensive testing of the integrated system to ensure seamless operation and reliable performance.

4.2 Prototype

Chapter 5

RESULTS AND DISCUSSION

5.1 Results

- **Sensor Accuracy and Reliability:**
  - DHT22 sensor: Accurate readings for ambient temperature (±0.5°C) and humidity (±2%).
  - NTC thermistor: Accurate battery temperature readings (±1°C).
  - Analog sensors: Minimal deviation in voltage and current measurements.
- **Real-time Data Transmission and Visualization:**
  - Stable MQTT communication from ESP32 to Node-RED dashboard.
  - Accurate real-time data updates on the Node-RED dashboard.
  - Clear and readable real-time data and alerts on the OLED display.
- **System Integration and Performance:**
  - Successful integration and seamless operation of all components.
  - Accurate response to environmental changes and effective alert triggering.
- **Data Logging and Historical Analysis:**
  - Reliable data logging functionality for sensor data storage.
  - Effective historical data retrieval mechanism for performance evaluation.

5.2 Discussion

- **Accuracy and Reliability:**
  - Accurate and reliable sensor readings ensure effective real-time monitoring.
  - Robust integration of components highlights system reliability.
- **User Interface:**
  - Intuitive and accessible Node-RED dashboard and OLED display enhance user experience.
  - Real-time visualization and alerts improve monitoring and decision-making.
- **Areas for Improvement:**
  - Further calibration of analog sensors for improved accuracy.
  - Extended long-term testing in diverse environmental conditions.
  - Enhancements to user interface for better data accessibility.
  - Implementation of scalability features for supporting multiple charging stations and centralized monitoring.
- **Overall Impact:**
  - The Quilles system demonstrates significant potential for enhancing the safety and efficiency of EV charging stations.
  - Comprehensive real-time monitoring and alerting capabilities contribute to optimal charging conditions.

Chapter 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

- The Quilles system effectively monitors battery health for EV charging stations using an IoT-based approach.
- Integration of sensors, ESP32 microcontroller, MQTT communication, and Node-RED ensures accurate real-time data transmission and visualization.
- Sensor readings for ambient temperature, humidity, battery temperature, voltage, and current are accurate and reliable.
- The OLED display provides clear, real-time data and alerts, enhancing user experience.
- Node-RED dashboard offers comprehensive monitoring and historical data analysis.
- The system demonstrates robustness, reliability, and effectiveness in ensuring optimal charging conditions, contributing to the safety and efficiency of EV charging stations.

6.2 Future Scope of Work

- **Enhanced Sensor Calibration:**
    - Further calibration of analog sensors for improved voltage and current measurement accuracy.
- **Extended Testing:**
    - Conduct long-term testing in diverse and extreme environmental conditions to evaluate system reliability and durability.
- **User Interface Improvements:**
    - Enhance the user interface of the Node-RED dashboard and OLED display for better data accessibility and interpretation.
- **Scalability:**
    - Implement features to support multiple charging stations and centralized monitoring for larger deployments.
- **Advanced Data Analytics:**
    - Integrate advanced data analytics and machine learning algorithms for predictive insights and enhanced decision-making.
- **Mobile App Integration:**
    - Develop a mobile application for remote monitoring and control, providing greater flexibility and accessibility.
- **Integration with Other Systems:**
    - Explore integration with other smart grid and energy management systems to enhance overall functionality and efficiency of the charging infrastructure.

REFERENCES

WEB:

1. About ESP32, http://www.espressif.com
2. Node-RED, nodered.org
3. MQTT Protocol, www.hivemq.com/mqtt-protocol
4. DHT22 Sensor, learn.adafruit.com
5. OLED Display, www.adafruit.com
6. NTC Thermistor, www.ti.com
7. Analog Sensors, www.analog.com
8. ESP32 Setup and Programming, www.developer.ibm.com
9. Implementing MQTT with ESP32, www.emqx.io
10. Using Node-RED for Dashboard Visualization, nodered.org/docs

LIVE DEMO:

https://github.com/nav9v/quilles-pulse